Robinson George Arysseril

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Education

Vimal Jyothi Engineering College — Chemperi, Kerala
B.Tech in Artificial Intelligence and Data Science CGPA: 9.46 / 10 — *Sep. 2024 – May 2028*

- Minor in Electronics and Communications

Technical Skills

Machine Learning & AI: PyTorch, TensorFlow, Transformers, Sentence-Transformers, Scikit-Learn, Pandas, NumPy, Matplotlib, SciPy, LSTM, MinMax Scaling
General Purpose: Python, Java, JavaScript, TypeScript, MySQL
Systems & Embedded: C/C++, Rust, Zig, Xtensa Assembly
Web & UI: Tauri, Angular, Bootstrap, HTML/CSS
Developer Tools: Git, Make

Experience

Research Intern — May 2026 – Jun. 2026
Amrita School of Artificial Intelligence, Amrita Vishwa Vidyapeetham — *Coimbatore, India*

- Investigated and analyzed gender bias in AI systems under Dr. Premjith B, Assistant Professor, using modern NLP and machine learning techniques on the project *"Gender Bias Detection in AI Systems"*
- Evaluated multiple text embedding models for their effectiveness at representing linguistic features relevant to bias detection in news headlines
- Developed and benchmarked both classical machine learning models and deep learning architectures for bias classification and analysis tasks
- Applied prompt engineering techniques to assess and mitigate bias in Large Language Models and generative AI systems

Open Source Contributor — May 2023 – Present
Chrultrabook Project — *Remote*

- Maintained Chrultrabook Tools, a cross-platform system utility serving users running alternate operating systems on Chromebook hardware
- Provided technical support and troubleshooting for community members transitioning Chromebooks to Linux, Windows, and other operating systems
- Collaborated with project maintainers to improve documentation, hardware compatibility databases, and user-facing tooling

Projects

Gender Bias Detection in News Headlines
Python, PyTorch, Sentence-Transformers, Transformers, Scikit-Learn — May 2026 – Jun. 2026

- Built an NLP bias-classification pipeline over a large news-headline corpus, generating Sentence-BERT embeddings and training a custom feed-forward classifier jointly optimized with cross-entropy and contrastive losses under learnable uncertainty-based loss weighting
- Designed a paired-sampling dataset for contrastive representation learning and validated the classifier with stratified 5-fold cross-validation, reaching 97.3% peak validation accuracy and ~94% macro-averaged test accuracy
- Implemented a neutrosophic (truth/indeterminacy/falsity) membership scoring function over embedding-centroid cosine similarity for uncertainty-aware, soft bias classification
- Built a two-step self-correcting LLM prompting pipeline, combining few-shot reasoning with a verification cross-examination pass, to evaluate prompt-engineering-based bias detection with SmolLM2-1.7B-Instruct

Various Emulators | *Rust, C++, Zig* — Oct. 2024 – Present

- **Game Boy (DMG):** Cycle-accurate emulator implementing CPU (Sharp LR35902), PPU, memory controller, and MBC1/MBC3 cartridge mappers with scanline-based rendering, tile caching, sprite compositing, window layer support, and OAM DMA timing

- **Intel 8080 / Space Invaders :** CPU emulator with complete instruction set support, accurate flag operations, and memory-mapped I/O with interrupt-driven timing synchronized to cycle counts
- **XO-CHIP :** Extended instruction set in Zig supporting dual display planes, 64KB memory addressing via 16-bit index register extension, and CHIP-8/SCHIP compatibility
- **CHIP-8 :** Interpreter in both Rust and C++ with full opcode coverage, stack operations, timers, input handling, and a framebuffer graphics system

Chrultrabook Tools | *Rust, Tauri, Angular, TypeScript* May 2023 – Present

- Cross-platform desktop application with real-time hardware sensor polling, fan curve control, and diagnostic data aggregation across Windows, macOS, and Linux; over 45,000 downloads
- Rust backend for direct hardware access enabling low-latency, high-frequency sensor reads critical for accurate runtime telemetry
- Modular architecture supporting multiple Chromebook hardware generations with minimal code changes for new device integration

One-Click Driver Installer & Hardware Database | *Rust, JSON, SQL* Sep. 2024 – Oct. 2024

- Structured hardware compatibility database cataloging 200+ Chromebook models with chipset and driver metadata; dynamic runtime fetch eliminates rebuilds when new models are added
- Automated driver installation CLI with over 5,000 downloads, demonstrating end-to-end data pipeline design from structured database to user-facing tooling